LECTURE 6. DIFFERENTIAL FORMS

6.1. **Derivations and Kähler differentials.** Fix k a field of characteristic zero and A/k be a k-algebra. Let M be a A-module. A *k-derivation on A with values in M* is a k-linear morphism

$$d_M : (A, +) \to (M, +)$$

satisfying the Leibniz rule

$$\forall x, y \in A, d_M(x \cdot y) = d_M(x) \cdot y + x \cdot d_M(y).$$

Notice that k-linearity of d_M is equivalent to the requirement that $\forall x \in k, d_M(x) = 0$.

Lemma 6.1. *There exists a (unique) pair $(\Omega^1(A/k), d)$ where $\Omega^1(A/k)$ is a A-module and d is a derivation on A with values in $\Omega^1(A/k)$ satisfying the following universal property:*

> *for every pair (M, d_M) as above, there exists a unique morphism of A-modules $\Omega^1(A/k) \to M$ such that:*

$$\begin{tikzcd} A \arrow[r, "d"] \arrow[d, "d_M"] & \Omega^1(A/k) \arrow[dl, dotted, "\exists!"] \\ M & \end{tikzcd} \quad \textit{commutes.}$$

The pair $(\Omega^1(A/k), d)$ is called the module of *one-forms on A/k* or *the module of k-differentials on A*.

Proof. Consider $E = \mathrm{Span}_A\{\delta x \mid x \in A\}$ the free A-module whose basis is given by the symbols δx. Let $\mathcal{R}$ the submodule of E generated by all the elements of the form

$$\delta(x + y) - \delta x - \delta y, \delta(xy) - x\delta y - y\delta x \text{ and } \delta c$$

where x, y ranges over all the elements of A and c ranges over all the elements of k. Define

$$\Omega^1(A/k) = E/\mathcal{R} \text{ and } d : x \mapsto x = \overline{\delta x} \in \Omega^1(A/k).$$

By construction, $(\Omega^1(A/k), d)$ is a module with a derivation. Now consider (M, d_M) another module with a derivation. The function $\delta x \mapsto d_M x$ extends uniquely to a morphism of A-module $\phi : E \to M$. Since d_M is a derivation, we have $\phi(\mathcal{R}) \subset \mathrm{Ker}(\phi)$ so that ϕ factors through

$$\overline{\phi} : E/\mathcal{R} = \Omega^1(A/k) \to M.$$

By construction, for every $x \in K$,

$$\overline{\phi}(\delta x) = \phi(\delta x) = d_M x$$

which shows that the diagram in the lemma commutes. Uniqueness follows from the fact that $\Omega^1(A/k)$ is generated by the dx with $x \in K$ (exercise). $\square$

The Leibniz rule together with additivity are sufficient properties to ensure that the differential calculus happens "as intended". For example,

Exercise 6.2. *Let $P(X_1, \dots, X_n) \in k[X_1, \dots, X_n]$ and $z_1, \dots, z_n \in A$. Then*

$$d(P(z_1, \dots, z_n)) = \sum_{i=1}^{n} \frac{\partial P}{\partial X_i}(z_1, \dots, z_n) \cdot dz_i. \tag{1}$$

Exercise 6.3. *Let R and S be two rings and $f : R \to S$. Adapt the previous framework to define a module of Kähler differentials attached to the morphism f which in the case where R is a field generalizes the previous framework.*

6.2. **The Lie derivative and connections.** Assume now that k is a differential field and A is a differential ring over k. Denote by ∂ the derivation on A.

Lemma 6.4. *There is a unique* additive map $\mathcal{L}_\partial : \Omega^1(K/k) \to \Omega^1(K/k)$ *satisfying:*

(1) the Leibniz rule*: for all $a \in K$ and all $\omega \in \Omega^1(K/k)$*

$$\mathcal{L}_\partial(a \cdot \omega) = \partial(a) \cdot \omega + a \cdot \mathcal{L}_\partial(\omega).$$

(2) the chain rule*: $d \circ \partial = \mathcal{L}_\partial \circ d$.*

Proof. The uniqueness part is clear by (2) since the one-forms of the form dx and $x \in A$ generate $\Omega^1(A/k)$. Indeed, if $\omega = \sum_{i=1}^n \lambda_i \cdot dx_i$ then

$$\mathcal{L}_\partial(\omega) = \sum_{i=1}^{n} \mathcal{L}_\partial(\lambda_i \cdot dx_i) = \sum_{i=1}^{n} \Big(\partial(\lambda_i) \cdot dx_i + \lambda_i \cdot d(\partial(x_i))\Big).$$

To show existence, recall the construction of $\Omega^1(A/k) = E/\mathcal{R}$ from Lemma 6.1 and consider the map defined by the previous formula

$$L_\partial : \sum_{i=1}^{n} \lambda_i \cdot \delta x_i \mapsto \sum_{i=1}^{n} \Big(\partial(\lambda_i) \cdot \delta x_i + \lambda_i \cdot \delta(\partial(x_i))\Big)$$

which is a additive map from E to E. To show that it descends to $\Omega^1(A/k)$, we need to show that $L_\delta(\mathcal{R}) \subset \mathcal{R}$. Consider $x, y \in A$

$$\begin{aligned} L_\partial(\delta(xy) - x\delta y - y\delta x) &= \delta(\partial(xy)) - \partial(x)\delta(y) - y\delta(\partial(x)) - \partial(y)\delta(x) - x\delta(\partial(y)) \\ &= \Big(\delta(x\partial(y) + y\partial(x)) - \delta(x\partial(y)) - \delta(y\partial(x))\Big) \\ &+ \Big(\delta(x\partial(y)) - x\delta(\partial(y)) - \partial(x)\delta(y)\Big) \\ &+ \Big(\delta(y\partial(x)) - y\delta(\partial(x)) - \partial(y)\delta(x)\Big) \in \mathcal{R} \end{aligned}$$

as it is the sum of three elements from $\mathcal{R}$. The other generators of $\mathcal{R}$ satisfy (easier) identities (exercise). It follows that L_∂ factors through an additive map $\mathcal{L}_\partial : E/\mathcal{R} \to E/\mathcal{R}$ satisfying the required properties. □

Definition 6.5. Let A/k be a differential k-algebra. Let M be a A-module. A *connection* on M is a k-linear map $\nabla_M : M \to M$ satisfying the Leibniz rule (1).

By the previous lemma, the Lie-derivative

$$\mathcal{L}_\partial : \Omega^1(A/k) \to \Omega^1(A/k)$$

defines a connection on $\Omega^1(A/k)$.

6.3. **Linear independence of one-forms.** We will often consider the case where $A = K$ is a differential field extension of k.

Exercise 6.6 (Localisation)**.** *Let S be multiplicative subset of A show that*

$$\Omega^1(S^{-1}A/k) = S^{-1}\Omega^1(A/k).$$

Theorem 6.7. *Let K/k be an extension of* fields *of characteristic zero and $(z_\alpha \mid \alpha \in A)$ a collection of elements from K. Then*

$(z_\alpha \mid \alpha \in A)$ *is a transcendence basis of $K/k \Leftrightarrow (dz_\alpha \mid \alpha \in A)$ is a K-linear basis of $\Omega^1(K/k)$.*

In particular, $\mathrm{td}(K/k) = \mathrm{ldim}_K(\Omega^1(K/k))$.

Proof. $\Rightarrow$ Assume that $(z_\alpha \mid \alpha \in A)$ is a transcendence basis of K/k. To see that the $(dz_\alpha \mid \alpha \in A)$ generates $\Omega^1(K/k)$, it is enough to see that its K-linear span contains all the elements of the form dx for $x \in K$. By assumption and characteristic zero, any such element satisfies a polynomial relation of the form

$$P(x, \overline{z_\alpha}) = 0 \text{ and } \frac{\partial P}{\partial X_0}(x, \overline{z_\alpha}) \neq 0 \text{ where } P \in k[X_0, \ldots, X_n] \text{ and } \overline{z_\alpha} = z_{\alpha_1}, \ldots, z_{\alpha_n}$$

Equation (1) implies that

$$0 = dP(x, \overline{z_\alpha}) = \frac{\partial P}{\partial X_0}(x, \overline{z_\alpha})dx + \sum_{i=1}^{n} \frac{\partial P}{\partial X_i}(x, \overline{z_\alpha})dz_{\alpha_i}$$

which shows that dx is in the K-linear span of the $(dz_\alpha \mid \alpha \in A)$. It remains to show that the $(dz_\alpha \mid \alpha \in A)$ are K-linearly independent in $\Omega^1(K/k)$.

Claim. *For every $\alpha \in K$, there is a (unique) derivation $\partial_\alpha \in \mathrm{Der}(K/k)$ on K trivial on k such that*

$$\partial_\alpha(\alpha) = 0 \textit{ and } \partial_\alpha(\beta) = 0 \textit{ for every } \beta \neq \alpha \in A$$

Proof of the claim. Decompose K/k as

$$k \subset L = k(z_\alpha \mid \alpha \in A) \subset K$$

The existence and uniqueness of ∂_α as a derivation of $\mathrm{Der}(L/k)$ follow from the presentation of L/k as a purely transcendental extension. The claim follows from the fact that any derivation on L extends uniquely to K since K/L is an algebraic extension. $\square$

Now consider a (finite) K-linear relation among the dz_α given as

$$\sum_{\alpha \in A} \lambda_\alpha \cdot dz_\alpha = 0 \text{ where } \lambda_\alpha \in K \text{ all but finitely many are zero}$$

and denote by ∂_α the derivation associated to z_α given by the previous claim. By the universal property of Lemma 6.1, we can find a K-linear form

$$\phi_\alpha : \Omega^1(K/k) \to K$$

with the property that $\phi_\alpha(dz) = \partial_\alpha(z)$ for all $z \in K$. It follows that for all $\beta \in A$,

$$0 = \phi_\beta(\sum_{\alpha \in A} \lambda_\alpha \cdot dz_\alpha) = \sum_{\alpha \in A} \lambda_\alpha \cdot \phi_\beta(dz_\alpha) = \lambda_\beta.$$

Hence, all the coefficients in the linear combination are trivial. We have therefore shown that the $(dz_\alpha \mid \alpha \in A)$ are K-linearly independent and hence form a K-basis of $\Omega^1(K/k)$.

$\Leftarrow$. For the converse, assume first for the sake of contradiction that the z_α are not algebraically independent over k. We can therefore find an algebraic relation of the form

$$P(z_\beta, \overline{z_\alpha}) = 0 \text{ and } \frac{\partial P}{\partial X_0}(z_\beta, \overline{z_\alpha}) \neq 0 \text{ where } P \in k[X_0, \ldots, X_n] \text{ and } \overline{z_\alpha} = z_{\alpha_1}, \ldots, z_{\alpha_n}$$

and the α_i are distinct from β. Using Equation 1, we obtain as previously

$$0 = dP(z_\beta, \overline{z_\alpha}) = \frac{\partial P}{\partial X_0}(z_\beta, \overline{z_\alpha})dz_\beta + \sum_{i=1}^{n} \frac{\partial P}{\partial X_i}(z_\beta, \overline{z_\alpha})dz_{\alpha_i}$$

which implies that $\frac{\partial P}{\partial X_0}(z_\beta, \overline{z_\alpha}) = 0$, a contradiction. We have therefore shown that the z_α are algebraically independent over k. To show that they form a transcendence basis of K/k, consider the decomposition

$$k \subset L = k(z_\alpha \mid \alpha \in A) \subset K$$

and assume for the sake of a contradiction that K/L is not an algebraic extension. We can therefore find a derivation ∂ on K which is trivial on L but not on K; say $\partial(x) \neq 0$ for some element $x \in K$. On the one hand, we can write

$$dx = \sum_{\alpha \in A} \lambda_\alpha \cdot dz_\alpha$$

where the right hand side is a finite sum. On the other, we have a K-linear form $\phi : \Omega^1(K/k) \to K$ such that $\phi(dz) = \partial(z)$ for all $z \in K$ by the universal property of Lemma 6.1. We conclude that

$$0 \neq \partial(x) = \phi(dx) = \sum_{\alpha \in A} \lambda_\alpha \cdot \phi(dz_\alpha) = 0$$

which is a contradiction. This finishes the proof of the theorem. □

6.4. **Closed rational one-forms.** Let K/k be a field extension.

Corollary 6.8. *Assume that K/k has finite transcendence degree then*

$$\Omega^1(K/k) = \mathrm{Der}(K/k)^* \text{ and } d : x \mapsto (\partial \mapsto \partial(x)).$$

Proof. The map d is well-defined since the evaluation of a derivation at a point is K-linear with respect to the derivation. Furthermore, k-linearly of d is obvious since if $x \in k$ then $\partial(x) = 0$ for all $x \in k$. Finally, for $x, y \in K$ and $\partial \in \mathrm{Der}(K/k)$

$$d(xy)(\partial) = \partial(xy) = \partial(x)y + x\partial(y) = (ydx + xdy)(\partial)$$

and the Leibniz rule follows. It follows by the universal property that

$$\phi : \Omega^1(K/k) \to \mathrm{Der}(K/k)^*$$

sending the differential d on $\Omega^1(K/k)$ to the newly defined d on $\mathrm{Der}(K/k)^*$. Pick $z_1, \ldots, z_n$ a transcendence basis of K/k. By Theorem 6.7,

$$\Omega^1(K/k) = Kdz_1 \oplus \ldots \oplus Kdz_n.$$

Using the claim of the previous theorem, we also obtain n derivations $\partial_1, \ldots, \partial_n$ on K such that $\partial_i(z_j) = 0$ if $i \neq j$ and is equal to 1 if $i = j$. An easy exercise shows that

$$\mathrm{Der}(K/k) = K\partial_1 \oplus \ldots \oplus K\partial_n.$$

Finally, since $dz_i(\partial_j) = \partial_j(z_i) = \begin{cases} 0 \text{ if } i \neq j \\ 1 \text{ if } i = j \end{cases}$, the morphism ϕ sends the basis defined dz_i to the dual of the basis defined by ∂_i. It follows that ϕ is an isomorphism. □

Definition 6.9. Assume that K/k has finite transcendence degree. A *two-form* ω on a field extension K/k is a K-bilinear map

$$\omega : \mathrm{Der}(K/k) \times \mathrm{Der}(K/k) \to K$$

which is alternating in the sense that for every $\partial_1, \partial_2 \in \mathrm{Der}(K/k)$, $\omega(\partial_1, \partial_2) = -\omega(\partial_2, \partial_1)$.

We denote by $\Omega^2(K/k)$ the K-vector space of 2-forms of dimension $\binom{n}{2}$. It is well-known (see any reference in linear algebra) that we have (anti commutative) product denoted

$$\wedge : \Omega^1(K/k) \times \Omega^1(K/k) \to \Omega^2(K/k)$$

given by the formula

$$(\omega_1 \wedge \omega_2)(\partial_1, \partial_2) = \omega_1(\partial_1) \cdot \omega_2(\partial_2) - \omega_1(\partial_2) \cdot \omega_2(\partial_1).$$

Lemma 6.10. *Let K/k be a finitely generated extension of fields and $\omega \in \Omega^1(K/k)$. Then the map*

$$d\omega : (\partial_1, \partial_2) \mapsto \partial_1(\omega(\partial_2)) - \partial_2(\omega(\partial_1)) - \omega([\partial_1, \partial_2])$$

where $[\partial_1, \partial_2] = \partial_1 \circ \partial_2 - \partial_2 \circ \partial_1$ is the Lie bracket of derivations in $\mathrm{Der}(K/k)$.

Proof. The definition of $d\omega$ shows that $d\omega$ is alternating. To show that it is bilinear, it is therefore enough to show that it is linear in the first variable. Consider $f \in K$ and $\partial_1, \partial_2 \in \mathrm{Der}(K/k)$.

$$\begin{aligned} d\omega(f\partial_1, \partial_2) &= f\partial_1(\omega(\partial_2)) - \partial_2(\omega(f\partial_1)) - \omega([f\partial_1, \partial_2]) \\ &= f\partial_1(\omega(\partial_2)) - \partial_2(f\omega(\partial_1)) - \omega(f[\partial_1, \partial_2] - \partial_2(f)\partial_1) \\ &= f\partial_1(\omega(\partial_2)) - \Big(f\partial_2(\omega(\partial_1)) + \partial_2(f)\omega(\partial_1)\Big) - \Big(f\omega(\partial_1, \partial_2)) - \partial_2(f)\omega(\partial_1)\Big) \\ &= f\partial_1(\omega(\partial_2)) - f\partial_2(\omega(\partial_1)) - f\omega(\partial_1, \partial_2)) = fd\omega(\partial_1, \partial_2) \end{aligned}$$

The proof of additivity is easier and left as an exercise. □

Definition 6.11. We say that a one-form $\omega \in \Omega^1(K/k)$ is *closed* if the two-form $d\omega \in \Omega^2(K/k)$ given by Lemma 6.10 is equal to zero.

References

[Har77] Robin Hartshorne. *Algebraic geometry.* Graduate Texts in Mathematics, No. 52. Springer-Verlag, New York-Heidelberg, 1977.

[PPR21] Dmitrii Pavlov, Gleb Pogudin, and Yury Razmyslov. From algebra to analysis: new proofs of theorems by ritt and seidenberg, 2021.

[Sei58] A. Seidenberg. Abstract differential algebra and the analytic case. *Proceedings of the American Mathematical Society*, 9(1):159–164, 1958.